Reporte Oficial de Mantenibilidad y Deuda Técnica

Análisis Estático y Financiero del Código Fuente (Arquitectura Limpia, SOLID, PHAME) — Actualizado: 18/10/2026

DEUDA TÉCNICA TOTAL (2)

12,281.72 hs

Equivalente a **76.76 meses** (4) sobre 136 archivos (20,450 LOC)

COSTO TOTAL DE REPARACIÓN (3)

$ 245,634.30

Calculado a razón de $20 USD / hora de desarrollo

ÍNDICE MANTENIBILIDAD PROMEDIO (1)

33.11 / 100

Interés Anual Base (5): **$1,400 USD** por módulo

VARIABLES GLOBALES (GLOSARIO)

- **LOC:** Líneas de código (sin blancos ni comentarios)
- **G:** Complejidad ciclomática de McCabe
- **K (Factor de Corrección):** 0.01
- **Salario por Hora:** $20.00 USD
- **Horas por Mes:** 160 horas laborables

✅ Plantilla Oficial de 7 Fórmulas Aplicada

1. `MI = max(0, ((171 - 25·ln(LOC) - 0.23·G)/171)·100)`
2. `Deuda (Hs) = max(0, (75 - MI)·LOC·K)`
3. `Costo = Deuda (Hs) × Salario/Hora`
4. `Meses = Deuda (Hs) / Horas_Mes`
5. `Interés Anual = 10 × (14 - 7) × Salario`
6. `Payback = Costo / Interés Anual`
7. `ROI = 100 × (((Interés Anual × 2) - Costo) / Costo)`

Archivos con Mayor Deuda Técnica (Top 8 Prioritarios)

Código Fuente Propio (Sin venv / node_modules)

Archivo	LOC	G	1. MI	2. Deuda	3. Costo ($)	4. Meses	5. Interés	6. Payback	7. ROI
`js/accounts.ts`	1268	462	**0.00**	951.00 hs	$19,020.00	5.94 m	$1,400	13.59 a	**-85.3%**
`js/shopping.ts`	911	265	**0.00**	683.25 hs	$13,665.00	4.27 m	$1,400	9.76 a	**-79.5%**
`js/services/scannerService.ts`	827	312	**0.00**	620.25 hs	$12,405.00	3.88 m	$1,400	8.86 a	**-77.4%**
`js/goals.ts`	757	204	**0.00**	567.75 hs	$11,355.00	3.55 m	$1,400	8.11 a	**-75.3%**
`js/budgets.ts`	695	173	**0.00**	521.25 hs	$10,425.00	3.26 m	$1,400	7.45 a	**-73.1%**
`js/supermercados.ts`	678	170	**0.00**	508.50 hs	$10,170.00	3.18 m	$1,400	7.26 a	**-72.5%**
`mobile/lib/screens/shopping_screen.dart`	573	82	**0.00**	429.75 hs	$8,595.00	2.69 m	$1,400	6.14 a	**-67.4%**
`js/reports.ts`	468	74	**0.16**	350.27 hs	$7,005.31	2.19 m	$1,400	5.00 a	**-60.0%**

** Nota: Este reporte audita exclusivamente los archivos fuente propios del repositorio (directorios api/, js/ y mobile/lib/), excluyendo librerías de terceros (venv, fluxo_venv, node_modules) para reflejar el costo operativo real.*

Métricas Detalladas por Archivo (1/5)

Todas las 7 Métricas Oficiales

Archivo	LOC	G	1. MI	2. Deuda	3. Costo ($)	4. Meses	5. Interés	6. Payback	7. ROI
api/wallet_adapters/mercadopago_adapter.py	360	65	**5.20**	251.27 hs	$5,025.37	1.57 m	$1,400	3.59 a	**-44.3%**
api/services/sepa/comparador.py	303	45	**10.41**	195.70 hs	$3,913.96	1.22 m	$1,400	2.80 a	**-28.5%**
api/services/recommendation.py	292	63	**8.53**	194.08 hs	$3,881.68	1.21 m	$1,400	2.77 a	**-27.9%**
api/routers/wallets.py	265	46	**12.24**	166.32 hs	$3,326.39	1.04 m	$1,400	2.38 a	**-15.8%**
api/routers/accounts.py	235	36	**15.34**	140.20 hs	$2,804.05	0.88 m	$1,400	2.00 a	**-0.1%**
api/tests/test_sepa.py	240	1	**19.74**	132.63 hs	$2,652.52	0.83 m	$1,400	1.89 a	**5.6%**
api/services/auth_service.py	205	46	**15.99**	120.97 hs	$2,419.37	0.76 m	$1,400	1.73 a	**15.7%**
api/services/shopping/generic_extractor.py	184	84	**12.46**	115.07 hs	$2,301.47	0.72 m	$1,400	1.64 a	**21.7%**
api/services/shopping_service.py	191	51	**16.35**	112.02 hs	$2,240.33	0.70 m	$1,400	1.60 a	**25.0%**
api/main.py	191	46	**17.03**	110.73 hs	$2,214.64	0.69 m	$1,400	1.58 a	**26.4%**
api/services/ai_service.py	185	44	**17.76**	105.89 hs	$2,117.85	0.66 m	$1,400	1.51 a	**32.2%**
api/routers/sepa.py	194	9	**21.77**	103.26 hs	$2,065.17	0.65 m	$1,400	1.48 a	**35.6%**
api/models.py	187	1	**23.39**	96.52 hs	$1,930.31	0.60 m	$1,400	1.38 a	**45.1%**
api/services/shopping/mercadolibre_extractor.py	162	75	**15.53**	96.34 hs	$1,926.76	0.60 m	$1,400	1.38 a	**45.3%**
api/infrastructure/repositories/sqlalchemy_pri…	178	24	**21.01**	96.09 hs	$1,921.87	0.60 m	$1,400	1.37 a	**45.7%**
api/seed_dummy.py	180	13	**22.33**	94.80 hs	$1,896.08	0.59 m	$1,400	1.35 a	**47.7%**
api/alembic/versions/4faf7655fc78_baseline_mig…	181	1	**23.86**	92.56 hs	$1,851.12	0.58 m	$1,400	1.32 a	**51.3%**
api/services/sepa/loader.py	166	30	**21.23**	89.26 hs	$1,785.22	0.56 m	$1,400	1.28 a	**56.8%**
api/services/account_service.py	163	36	**20.69**	88.53 hs	$1,770.58	0.55 m	$1,400	1.26 a	**58.1%**
api/services/goal_service.py	158	47	**19.66**	87.43 hs	$1,748.62	0.55 m	$1,400	1.25 a	**60.1%**
api/services/transaction_service.py	126	27	**25.66**	62.17 hs	$1,243.30	0.39 m	$1,400	0.89 a	**125.2%**
api/tests/test_accounts.py	131	5	**28.05**	61.50 hs	$1,230.02	0.38 m	$1,400	0.88 a	**127.6%**
api/routers/auth.py	125	15	**27.39**	59.51 hs	$1,190.17	0.37 m	$1,400	0.85 a	**135.3%**
api/schemas.py	117	1	**30.24**	52.37 hs	$1,047.31	0.33 m	$1,400	0.75 a	**167.4%**
api/tests/test_goals.py	110	1	**31.15**	48.24 hs	$964.81	0.30 m	$1,400	0.69 a	**190.2%**
api/services/sepa/promociones.py	95	11	**31.94**	40.90 hs	$818.08	0.26 m	$1,400	0.58 a	**242.3%**
api/services/sepa/downloader.py	81	9	**34.54**	32.77 hs	$655.40	0.20 m	$1,400	0.47 a	**327.2%**
api/services/sepa/ingestion.py	79	4	**35.58**	31.14 hs	$622.82	0.19 m	$1,400	0.44 a	**349.6%**

Métricas Detalladas por Archivo (2/5)

Todas las 7 Métricas Oficiales

Archivo	LOC	G	1. MI	2. Deuda	3. Costo ($)	4. Meses	5. Interés	6. Payback	7. ROI
api/application/use_cases/sepa_use_cases.py	74	11	**35.60**	29.16 hs	$583.19	0.18 m	$1,400	0.42 a	**380.1%**
api/security.py	73	15	**35.26**	29.01 hs	$580.26	0.18 m	$1,400	0.41 a	**382.5%**
api/tests/test_recommendation.py	65	13	**37.22**	24.56 hs	$491.11	0.15 m	$1,400	0.35 a	**470.1%**
api/tests/test_budgets.py	64	3	**38.79**	23.17 hs	$463.44	0.14 m	$1,400	0.33 a	**504.2%**
api/infrastructure/dependencies.py	60	1	**40.01**	21.00 hs	$419.92	0.13 m	$1,400	0.30 a	**566.8%**
api/wallet_adapters/plaid_adapter.py	57	2	**40.62**	19.60 hs	$391.91	0.12 m	$1,400	0.28 a	**614.5%**
api/routers/ai.py	56	8	**40.07**	19.56 hs	$391.17	0.12 m	$1,400	0.28 a	**615.8%**
api/domain/entities/product.py	57	1	**40.76**	19.52 hs	$390.38	0.12 m	$1,400	0.28 a	**617.3%**
api/wallet_adapters/belvo_adapter.py	56	2	**40.88**	19.11 hs	$382.13	0.12 m	$1,400	0.27 a	**632.7%**
api/infrastructure/repositories/sqlalchemy_use…	52	7	**41.29**	17.53 hs	$350.57	0.11 m	$1,400	0.25 a	**698.7%**
api/routers/shopping.py	52	1	**42.10**	17.11 hs	$342.17	0.11 m	$1,400	0.24 a	**718.3%**
api/wallet_adapters/base_adapter.py	49	2	**42.83**	15.76 hs	$315.24	0.10 m	$1,400	0.23 a	**788.2%**
api/routers/goals.py	47	3	**43.31**	14.90 hs	$297.91	0.09 m	$1,400	0.21 a	**839.9%**
api/services/sepa/geo.py	44	6	**43.87**	13.70 hs	$273.96	0.09 m	$1,400	0.20 a	**922.1%**
api/infrastructure/repositories/sqlalchemy_wal…	44	1	**44.54**	13.40 hs	$268.04	0.08 m	$1,400	0.19 a	**944.6%**
api/routers/transactions.py	42	1	**45.22**	12.51 hs	$250.14	0.08 m	$1,400	0.18 a	**1019.4%**
api/alembic/env.py	40	12	**44.45**	12.22 hs	$244.36	0.08 m	$1,400	0.17 a	**1045.8%**
api/services/sepa/cleaner.py	40	11	**44.59**	12.16 hs	$243.28	0.08 m	$1,400	0.17 a	**1050.9%**
api/services/ocr.py	40	5	**45.40**	11.84 hs	$236.83	0.07 m	$1,400	0.17 a	**1082.3%**
api/services/budget_service.py	39	7	**45.50**	11.51 hs	$230.12	0.07 m	$1,400	0.16 a	**1116.8%**
api/database.py	37	15	**45.19**	11.03 hs	$220.58	0.07 m	$1,400	0.16 a	**1169.4%**
api/infrastructure/repositories/sqlalchemy_tra…	38	1	**46.68**	10.76 hs	$215.20	0.07 m	$1,400	0.15 a	**1201.1%**
api/infrastructure/repositories/sqlalchemy_goa…	36	1	**47.47**	9.91 hs	$198.18	0.06 m	$1,400	0.14 a	**1312.9%**
api/ports/repositories/price_repository_port.py	34	1	**48.31**	9.07 hs	$181.49	0.06 m	$1,400	0.13 a	**1442.8%**
api/services/shopping/extractor_registry.py	29	8	**49.69**	7.34 hs	$146.77	0.05 m	$1,400	0.10 a	**1807.7%**
api/routers/ocr.py	27	5	**51.14**	6.44 hs	$128.83	0.04 m	$1,400	0.09 a	**2073.4%**
api/infrastructure/repositories/sqlalchemy_acc…	26	1	**52.23**	5.92 hs	$118.39	0.04 m	$1,400	0.08 a	**2265.0%**
api/routers/budgets.py	26	1	**52.23**	5.92 hs	$118.39	0.04 m	$1,400	0.08 a	**2265.0%**

Métricas Detalladas por Archivo (3/5)

Todas las 7 Métricas Oficiales

Archivo	LOC	G	1. MI	2. Deuda	3. Costo ($)	4. Meses	5. Interés	6. Payback	7. ROI
api/infrastructure/repositories/sqlalchemy_bu…	23	1	**54.02**	4.82 hs	$96.49	0.03 m	$1,400	0.07 a	**2802.0%**
api/application/use_cases/auth_use_cases.py	22	1	**54.67**	4.47 hs	$89.43	0.03 m	$1,400	0.06 a	**3030.9%**
api/ports/repositories/goal_repository_port.py	21	1	**55.35**	4.13 hs	$82.51	0.03 m	$1,400	0.06 a	**3293.6%**
api/ports/repositories/transaction_repository…	21	1	**55.35**	4.13 hs	$82.51	0.03 m	$1,400	0.06 a	**3293.6%**
api/ports/repositories/wallet_repository_port…	17	1	**58.44**	2.81 hs	$56.29	0.02 m	$1,400	0.04 a	**4874.3%**
api/ports/repositories/account_repository_por…	15	1	**60.27**	2.21 hs	$44.18	0.01 m	$1,400	0.03 a	**6238.0%**
api/domain/entities/user.py	14	1	**61.28**	1.92 hs	$38.41	0.01 m	$1,400	0.03 a	**7190.1%**
api/ports/repositories/budget_repository_port…	13	1	**62.37**	1.64 hs	$32.85	0.01 m	$1,400	0.02 a	**8424.2%**
api/ports/repositories/user_repository_port.py	12	1	**63.54**	1.38 hs	$27.51	0.01 m	$1,400	0.02 a	**10077.2%**
api/domain/exceptions.py	10	1	**66.20**	0.88 hs	$17.60	0.01 m	$1,400	0.01 a	**15812.7%**
api/run_ingesta.py	10	1	**66.20**	0.88 hs	$17.60	0.01 m	$1,400	0.01 a	**15812.7%**
api/services/shopping/__init__.py	10	1	**66.20**	0.88 hs	$17.60	0.01 m	$1,400	0.01 a	**15812.7%**
api/services/sepa/config.py	9	1	**67.74**	0.65 hs	$13.06	0.00 m	$1,400	0.01 a	**21333.3%**
api/ports/extractors/product_extractor_port.py	6	1	**73.67**	0.08 hs	$1.60	0.00 m	$1,400	0.00 a	**175362.8%**
api/domain/entities/__init__.py	3	1	**83.80**	0.00 hs	$0.00	0.00 m	$1,400	0.00 a	**100.0%**
api/wallet_adapters/__init__.py	5	1	**76.34**	0.00 hs	$0.00	0.00 m	$1,400	0.00 a	**100.0%**
js/accounts.ts	1268	462	**0.00**	951.00 hs	$19,020.00	5.94 m	$1,400	13.59 a	**-85.3%**
js/shopping.ts	911	265	**0.00**	683.25 hs	$13,665.00	4.27 m	$1,400	9.76 a	**-79.5%**
js/services/scannerService.ts	827	312	**0.00**	620.25 hs	$12,405.00	3.88 m	$1,400	8.86 a	**-77.4%**
js/goals.ts	757	204	**0.00**	567.75 hs	$11,355.00	3.55 m	$1,400	8.11 a	**-75.3%**
js/budgets.ts	695	173	**0.00**	521.25 hs	$10,425.00	3.26 m	$1,400	7.45 a	**-73.1%**
js/supermercados.ts	678	170	**0.00**	508.50 hs	$10,170.00	3.18 m	$1,400	7.26 a	**-72.5%**
js/reports.ts	468	74	**0.16**	350.27 hs	$7,005.31	2.19 m	$1,400	5.00 a	**-60.0%**
js/services/shoppingService.ts	429	127	**0.00**	321.75 hs	$6,435.00	2.01 m	$1,400	4.60 a	**-56.5%**
js/transactions.ts	362	147	**0.00**	271.50 hs	$5,430.00	1.70 m	$1,400	3.88 a	**-48.4%**
js/shoppingList.ts	326	53	**8.27**	217.55 hs	$4,350.95	1.36 m	$1,400	3.11 a	**-35.6%**
js/scanner.ts	310	75	**6.04**	213.76 hs	$4,275.26	1.34 m	$1,400	3.05 a	**-34.5%**
js/import.ts	302	62	**8.18**	201.81 hs	$4,036.23	1.26 m	$1,400	2.88 a	**-30.6%**

Métricas Detalladas por Archivo (4/5)

Todas las 7 Métricas Oficiales

Archivo	LOC	G	1. MI	2. Deuda	3. Costo ($)	4. Meses	5. Interés	6. Payback	7. ROI
js/services/accountService.ts	244	84	**8.33**	162.67 hs	$3,253.31	1.02 m	$1,400	2.32 a	**-13.9%**
js/services/shoppingListService.ts	233	32	**16.00**	137.46 hs	$2,749.29	0.86 m	$1,400	1.96 a	**1.8%**
js/ui.ts	221	58	**13.28**	136.40 hs	$2,728.10	0.85 m	$1,400	1.95 a	**2.6%**
js/ai.ts	206	50	**15.38**	122.81 hs	$2,456.26	0.77 m	$1,400	1.75 a	**14.0%**
js/profile.ts	182	64	**15.31**	108.64 hs	$2,172.72	0.68 m	$1,400	1.55 a	**28.9%**
js/services/goalService.ts	171	44	**18.91**	95.91 hs	$1,918.23	0.60 m	$1,400	1.37 a	**46.0%**
js/services/sepaService.ts	165	33	**20.91**	89.24 hs	$1,784.87	0.56 m	$1,400	1.27 a	**56.9%**
js/app.ts	153	50	**19.73**	84.56 hs	$1,691.25	0.53 m	$1,400	1.21 a	**65.6%**
js/auth.ts	158	30	**21.95**	83.82 hs	$1,676.37	0.52 m	$1,400	1.20 a	**67.0%**
js/types.ts	126	33	**24.86**	63.18 hs	$1,263.64	0.39 m	$1,400	0.90 a	**121.6%**
js/components/chatbot.ts	102	13	**30.63**	45.25 hs	$905.05	0.28 m	$1,400	0.65 a	**209.4%**
js/utils/utils.ts	83	29	**31.50**	36.11 hs	$722.16	0.23 m	$1,400	0.52 a	**287.7%**
js/services/authService.ts	84	20	**32.53**	35.67 hs	$713.46	0.22 m	$1,400	0.51 a	**292.5%**
js/store/store.ts	68	13	**36.56**	26.14 hs	$522.75	0.16 m	$1,400	0.37 a	**435.6%**
js/services/transactionService.ts	60	9	**38.93**	21.64 hs	$432.83	0.14 m	$1,400	0.31 a	**546.9%**
js/services/budgetService.ts	54	7	**40.74**	18.50 hs	$370.01	0.12 m	$1,400	0.26 a	**656.7%**
js/api/apiClient.ts	39	14	**44.56**	11.87 hs	$237.46	0.07 m	$1,400	0.17 a	**1079.1%**
js/api/authApi.ts	40	1	**45.93**	11.63 hs	$232.52	0.07 m	$1,400	0.17 a	**1104.2%**
js/services/cameraService.ts	39	8	**45.36**	11.56 hs	$231.17	0.07 m	$1,400	0.17 a	**1111.2%**
js/services/chatService.ts	30	7	**49.33**	7.70 hs	$154.00	0.05 m	$1,400	0.11 a	**1718.2%**
mobile/lib/screens/shopping_screen.dart	573	82	**0.00**	429.75 hs	$8,595.00	2.69 m	$1,400	6.14 a	**-67.4%**
mobile/lib/screens/supermercados_screen.dart	345	24	**11.34**	219.63 hs	$4,392.54	1.37 m	$1,400	3.14 a	**-36.3%**
mobile/lib/screens/transactions_screen.dart	248	26	**15.90**	146.57 hs	$2,931.50	0.92 m	$1,400	2.09 a	**-4.5%**
mobile/lib/screens/profile_screen.dart	245	23	**16.48**	143.38 hs	$2,867.54	0.90 m	$1,400	2.05 a	**-2.4%**
mobile/lib/screens/insights_screen.dart	240	20	**17.18**	138.76 hs	$2,775.19	0.87 m	$1,400	1.98 a	**0.9%**
mobile/lib/screens/accounts_screen.dart	204	22	**19.29**	113.65 hs	$2,272.94	0.71 m	$1,400	1.62 a	**23.2%**
mobile/lib/screens/goals_screen.dart	184	14	**21.88**	97.75 hs	$1,954.99	0.61 m	$1,400	1.40 a	**43.2%**
mobile/lib/screens/budgets_screen.dart	170	16	**22.76**	88.80 hs	$1,776.05	0.56 m	$1,400	1.27 a	**57.7%**

Métricas Detalladas por Archivo (5/5)

Todas las 7 Métricas Oficiales

Archivo	LOC	G	1. MI	2. Deuda	3. Costo ($)	4. Meses	5. Interés	6. Payback	7. ROI
mobile/lib/screens/scanner_screen.dart	166	12	**23.65**	85.24 hs	$1,704.84	0.53 m	$1,400	1.22 a	**64.2%**
mobile/lib/screens/login_screen.dart	162	16	**23.47**	83.48 hs	$1,669.64	0.52 m	$1,400	1.19 a	**67.7%**
mobile/lib/screens/dashboard_screen.dart	156	13	**24.42**	78.90 hs	$1,578.00	0.49 m	$1,400	1.13 a	**77.4%**
mobile/lib/providers/insights_provider.dart	145	36	**22.40**	76.27 hs	$1,525.44	0.48 m	$1,400	1.09 a	**83.6%**
mobile/lib/screens/main_layout.dart	152	9	**25.34**	75.48 hs	$1,509.64	0.47 m	$1,400	1.08 a	**85.5%**
mobile/lib/services/api_service.dart	134	28	**24.63**	67.50 hs	$1,349.97	0.42 m	$1,400	0.96 a	**107.4%**
mobile/lib/providers/auth_provider.dart	131	25	**25.36**	65.02 hs	$1,300.50	0.41 m	$1,400	0.93 a	**115.3%**
mobile/lib/screens/reports_screen.dart	125	11	**27.93**	58.84 hs	$1,176.72	0.37 m	$1,400	0.84 a	**137.9%**
mobile/lib/providers/supermercado_provider.dart	116	41	**24.99**	58.01 hs	$1,160.27	0.36 m	$1,400	0.83 a	**141.3%**
mobile/lib/models/supermercado_model.dart	96	25	**29.91**	43.29 hs	$865.78	0.27 m	$1,400	0.62 a	**223.4%**
mobile/lib/providers/shopping_provider.dart	61	20	**37.21**	23.05 hs	$461.04	0.14 m	$1,400	0.33 a	**507.3%**
mobile/lib/providers/accounts_provider.dart	48	16	**41.25**	16.20 hs	$323.99	0.10 m	$1,400	0.23 a	**764.2%**
mobile/lib/providers/transactions_provider.dart	48	16	**41.25**	16.20 hs	$323.99	0.10 m	$1,400	0.23 a	**764.2%**
mobile/lib/providers/goals_provider.dart	46	15	**42.01**	15.18 hs	$303.52	0.09 m	$1,400	0.22 a	**822.5%**
mobile/lib/providers/scanner_provider.dart	41	11	**44.23**	12.62 hs	$252.33	0.08 m	$1,400	0.18 a	**1009.7%**
mobile/lib/providers/budgets_provider.dart	39	13	**44.69**	11.82 hs	$236.41	0.07 m	$1,400	0.17 a	**1084.4%**
mobile/lib/providers/dashboard_provider.dart	38	12	**45.20**	11.32 hs	$226.44	0.07 m	$1,400	0.16 a	**1136.5%**
mobile/lib/main.dart	34	2	**48.18**	9.12 hs	$182.40	0.06 m	$1,400	0.13 a	**1435.1%**
mobile/lib/models/goal.dart	27	18	**49.39**	6.91 hs	$138.27	0.04 m	$1,400	0.10 a	**1925.0%**
mobile/lib/services/supermercado_service.dart	23	6	**53.35**	4.98 hs	$99.58	0.03 m	$1,400	0.07 a	**2711.8%**
mobile/lib/models/account.dart	20	8	**55.13**	3.97 hs	$79.49	0.02 m	$1,400	0.06 a	**3422.3%**
mobile/lib/models/budget.dart	20	6	**55.40**	3.92 hs	$78.42	0.02 m	$1,400	0.06 a	**3470.7%**
mobile/lib/models/transaction.dart	20	3	**55.80**	3.84 hs	$76.80	0.02 m	$1,400	0.05 a	**3545.7%**
mobile/lib/services/api_config.dart	3	1	**83.80**	0.00 hs	$0.00	0.00 m	$1,400	0.00 a	**100.0%**